Roll No. : .....

MID SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program: **M.Tech.** Semester: III

Name of the Course: Intrusion Detection System

Course Code: MCS 324

Time : 90 Minutes **Maximum Marks : 50**

Note:

i. Answer all the questions by choosing any one of the sub questions
ii. Each questions carries 10 marks

Q1	(10 Marks)	CO1
(a)	What is the Hansom taxonomy of attacks. Also Discuss the classification of attacks in detail depending on the main behavior of the attack.	
	OR	
(b)	Differentiate between Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) with examples. Also discuss the myths about IDS and IPS.	
Q2	(10 Marks)	CO1
(a)	A home security system can be compared to an Intrusion Prevention System (IPS): it not only detects intrusions but also automatically locks doors and windows to prevent further damage. In contrast, an Intrusion Detection System (IDS) in the same scenario would only send alerts without taking direct action. (a) Explain how this analogy applies to IDS and IPS. (b) Compare their advantages and disadvantages in terms of prevention capability, false positives, and administrative effort.	
	OR	
(b)	Intrusion Detection Systems (IDS) can be classified in multiple ways such as based on deployment and detection technique. Explain these classifications in detail with suitable examples.	
Q3	(10 Marks)	CO2
(a)	Web applications are common targets for attacks. Explain web-based attacks along with how an Intrusion Detection System (IDS) can detect these types of web attacks.	
	OR	
(b)	What do you understand by **anomaly detection** in Intrusion Detection Systems (IDS)? Explain how it works, and discuss its major benefits and challenges in real-world applications.	
Q4	(10 Marks)	CO2 & CO3
(a)	Explain the different approaches to data collection. Also, list and briefly describe commonly used tools for IDS data collection.	
	OR	

(b)	Analyze the role of Artificial Neural Networks (ANNs) in intrusion detection compared to traditional rule-based systems.	
Q5	(10 Marks)	CO3
(a)	Consider a dataset of network packets and failed login attempts. Demonstrate how fuzzy logic can be applied to classify traffic into safe or suspicious.	
	OR	
(b)	Demonstrate with the help of an example, how **SVM (support vector machine)** can be used to classify traffic as normal or anomalous.	